INTERNATIONAL MONETARY FUND

SENEGAL

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THIRD REVIEW UNDER THE POLICY COORDINATION INSTRUMENT AND REQUEST FOR MODIFICATION OF QUANTITATIVE TARGETS, AND REQUESTS FOR A STAND-BY ARRANGEMENT AND AN ARRANGEMENT UNDER THE STANDBY CREDIT FACILITY—DEBT SUSTAINABILITY ANALYSIS

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Prepared by the staffs of the International Monetary Fund and the International Development Association

Senegal: Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress:	*Moderate*[1]
Overall risk of debt distress	*Moderate*
Granularity in the risk rating	*Limited scope to absorb shocks*
Application of judgment	*No*

Senegal's faces a moderate risk of overall debt distress (external and public) under the current baseline scenario, with limited scope to absorb shocks in the near term. This rating is predicated on a gradual recovery from the COVID-19 pandemic, adherence to a prudent fiscal path (in line with regional convergence criteria) and strengthened debt management. The outlook for Senegal remains highly uncertain, primarily a reflection of uncertainty regarding the path of the COVID-19 pandemic, the pace of the vaccination campaign, and developments in key trading partners. Continued debt sustainability will need to be anchored in a prudent borrowing strategy that prioritizes concessional external borrowing and domestic regional financing in line with programmed financing needs, combined with continued efforts to strengthen debt management and contain fiscal risks.

[1] Senegal's debt carrying capacity is classified as strong (calculated based on the October 2020 WEO and 2019 World Bank's Country Policy and Institutional Assessment (CPIA) score). The applicable thresholds to public and publicly guaranteed external debt are: 55 percent for the Present Value (PV) of debt-to-GDP ratio, 240 percent for the PV of debt-to-exports ratio, 21 percent for the debt service-to-exports ratio, and 23 percent for the debt service-to-revenue ratio. The applicable benchmark for the PV of total public debt for strong debt carrying capacity is 70 percent of GDP.

DEBT COVERAGE

1. This DSA uses a broad definition of public debt. The assessment includes public and publicly-guaranteed debt held by (i) the central government, (ii) para-public entities, and (iii) state-owned enterprises (SOEs), combining data provided by the authorities with information compiled by IMF technical assistance (Text Table 1).[2] The DSA classifies external and domestic debt based on currency, given data constraints that prevent the use of a residency-based definition. In terms of contingent risks, the default financial sector shock of 5 percent of GDP is more than adequate to cover potential bank recapitalization needs discussed in the staff report, which are estimated to be less than 1 percent of GDP.

Text Table 1. Senegal: Coverage of Public Sector Debt and Design of the Contingent Liability Stress Test

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	X
3	Other elements in the general government	X
4	o/w: Social security fund	X
5	o/w: Extra budgetary funds (EBFs)	X
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	X

1	The country's coverage of public debt	The entire public sector, including SOEs		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	0.0	Already included in definition of government debt
4	PPP	35 percent of PPP stock	2.1	PPP capital stock of 6 percent of GDP is larger than 3 percent threshold
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)		7.1	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country

2. Efforts are underway to strengthen the quality and coverage of public debt data. The national debt committee (CNDP), chaired by the Minister of Finance, has been revitalized in the past year and is now reviewing all large public investment decisions, including those by SOEs. As part of the Fund-supported program, the authorities have engaged an audit firm to conduct an audit to assess whether the recording and communication of data on public debt (including the debt of the SOE sector) is accurate, exhaustive and in line with best practices. This should help improve the quality and timeliness of the data available to the authorities for planning and debt management purposes. Following some administrative delays, the audit is expected to be completed in the first half of 2021.

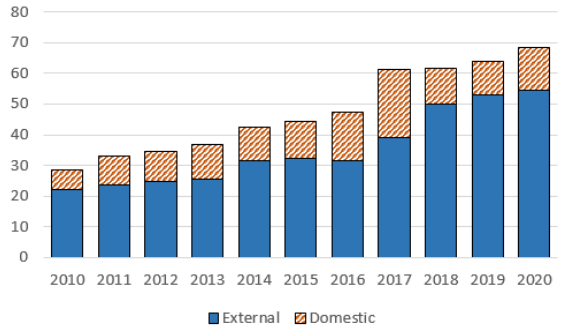
[2] The inclusion of para-public enterprises and SOEs began in 2017. The list of entities covered by the DSA is provided in the Technical Memorandum of Understanding. The 2018 public sector balance sheet was compiled with support from Fund TA. Previous DSAs erroneously indicated that the social security system was not covered by the debt perimeter.

BACKGROUND

3. Public sector debt levels have risen significantly in recent years. Increases in public investment have driven a substantial increase in public spending—particularly in 2017–18—linked in part to Senegal's national development plan, the Plan Senegal Emergent (PSE), and financial support for large state-owned enterprises. In addition, the perimeter of public debt was expanded in 2017 to capture public and para-public enterprises, creating a structural break in the data (Text Figure 1).[3]

4. Senegal has successfully accessed the Eurobond market on several occasions: in 2011 (US$500 million), 2017 (US$1.1 billion), and 2018 (US$2.2 billion). These Eurobonds have increased Senegal's external non-concessional debt and represent the largest source of commercial debt service. They also helped create space in the regional bond market for other WAEMU members and supported the build-up of much-needed WAEMU pooled reserves.

Text Figure 1. Senegal: Public Sector Debt
(percent of GDP)



Sources: Senegalese authorities, IMF estimates and projections.

5. The COVID-19 pandemic has contributed to higher public sector external debt and total public debt. As of end-December 2020, public sector external debt stood at 54.1 percent of GDP, compared to a projected 51.9 percent at the time of the last full DSA in January 2020.[4] Total public sector debt reached 68.7 percent of GDP, about 7 percentage points higher than the pre-pandemic forecast. This largely reflects the impact of the COVID-19 shock, which affected both external and domestic demand and led to lower economic growth (1.5 percent of GDP compared to a pre-crisis forecast of 6.8 percent of GDP) and a higher public sector deficit (6.3 percent of GDP compared to a pre-crisis forecast of 3 percent of GDP). Other debt creating flows in 2020 include below-the-line financing for capital transfers ("compte de dépôt") and the COVID guarantee fund (1.3 percent of GDP).[5]

6. Both external and total public debt service remain substantial. The ratio of public external debt service to exports reached 24.2 percent in 2020, reflecting the decline in exports due to the COVID shock and the below-the-line financing items noted above. This ratio is expected to improve to around 17.5 percent in 2021 as the recovery takes hold and the country benefits from the G-20 debt service relief initiative (discussed below). The ratio of public external debt service to revenue rose from 11.2 percent in 2019 to 16.4 percent in 2020 and is projected to average about 15 percent over the medium term. Total public debt service absorbed about 24 percent of revenue and grants in 2020 and is expected to remain at

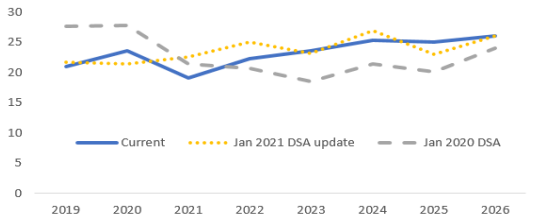
[3] Senegal is the only WAEMU member country to use this broader definition of public sector debt.

[4] IMF Country Report No. 20/11. The subsequent DSAs published in April 2020 (IMF Country Report No. 20/108) and January 2021 for the 2nd PCI review (IMF Country Report No. 21/18) were streamlined.

[5] The residual is largely explained by the fact that the LIC DSF calculates the real exchange rate effects on the automatic debt dynamics using the assumption that all external debt is denominated in USD. In fact, only about a third of Senegal's external public debt is denominated in USD.

that level on average over the medium term (Text Figure 2). The debt service projection includes substantial Eurobond repayments scheduled for 2024 and 2026. The authorities are considering a new Eurobond issuance in 2021 for liability management purposes, to smooth out the large amortization payments over time, but this has not yet been included in the baseline as the details have not yet been finalized.

Text Figure 2. Senegal: Public Debt Service to Revenue and Grants
(percent)



Sources: Senegalese authorities, IMF estimates and projections.

7. Senegal is participating in the G-20 Debt Service Suspension Initiative (DSSI). The DSSI is a net-present-value (NPV) neutral exercise intended to provide eligible members with liquidity relief to allow them to focus more resources on responding to the COVID-19 pandemic. The DSSI provided around CFAF 30 billion in debt service relief (0.2 percent of GDP), including CFAF 6 billion in interest (0.04 percent of GDP)) over May-December 2020. The G-20's decision to extend the initiative until the end of 2021 is expected to result in additional debt service relief of CFAF 91 billion (0.6 percent of GDP), including CFAF 19 billion in interest (0.1 percent of GDP). Repayments will occur over 2022–27 and average about CFAF 20 billion per year.

BASELINE SCENARIO

8. The macroeconomic assumptions underlying the projections are consistent with the proposed program baseline for the SCF/SBA. This includes higher growth in 2020 than anticipated at the time of the 2nd PCI review (January 2021), but also a more subdued recovery in 2021 as the 2nd wave of the COVID-19 pandemic takes a toll on economic activity. The DSA assumes that the authorities implement the macroeconomic policies and structural reforms outlined in the staff report, including a return to a central government fiscal deficit of 3 percent of GDP by 2023, consistent with the WAEMU convergence criterion. Long-run macroeconomic assumptions remain largely unchanged from the DSA update at the time of the 2nd PCI review. The main macroeconomic assumptions are as follows:

- **Real GDP Growth.** Senegal's real GDP growth rate is estimated to be 1.5 percent in 2020, with 3.7 percent projected for 2021, compared to -0.7 percent and 5.2 percent at the time of the 2nd PCI review, respectively. The growth rate is projected to peak at 10.8 percent of GDP in 2023 before declining to 6.1 percent in 2024, reflecting the incorporation of oil and gas production (see below). Over the long run, real GDP growth is projected to be about 5 percent.
- **Oil and gas**. The framework includes all three phases of the Sangomar (SNE) offshore oil field development and the Greater Tortue Ahmeyim (GTA) gas project, which continue to be developed despite the pandemic.[6] Production is expected to begin in the second half of 2023. The borrowing needs of Petrosen (the state-owned oil company) for both SNE and GTA are estimated at CFAF 1,228 billion from 2020 through 2037 (Text Table 2) although these estimates are subject to change as the projects are further developed. The Yakaar-Taranga project, which is also under development, has not been included in the DSA as it has not yet reached a final investment decision.

[6] GTA is exploited jointly with Mauritania.

Text Table 2. Senegal: Oil and Gas: Petrosen Borrowing Assumptions[1]

	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030-2037	Total
CFAF billions	131	195	163	124	102	65	90	62	58	55	182	1,228
% of GDP	0.24%	0.92%	1.30%	1.01%	0.68%	0.52%	0.31%	0.40%	0.26%	0.22%	0.19%	

1/ As the projects are under development, these figures are subject to change

Source: Senegalese authorities, IMF staff projections.

- **Inflation.** Inflation is projected to remain contained at about 2.2 percent in 2021 and to hover around 1.5 percent over the medium term.
- **Public sector deficit.** The public sector deficit—which includes the central government deficit and the net lending and borrowing of SOEs—is estimated to have reached 6.4 percent of GDP in 2020. This mainly reflects the higher fiscal deficits from the central government and large investments by Petrosen. The deficit is projected to be 5.4 percent of GDP in 2021 before gradually declining to around 3 percent over the medium term, reflecting improved government fiscal balances, lower financing needs in the hydrocarbon sector, and efforts to strengthen the financial performance of SOEs.
- **Revenues.** Relative to 2019, the central government's tax revenue declined by almost 1 percent of GDP due to the pandemic, down to 16.7 percent of GDP. Tax revenues are projected to gradually increase to about 20 percent of GDP in 2024 in line with the objectives of the medium-term revenue strategy and are expected to stay above that level over the medium term. This will be supported by the steady implementation of reforms to support revenue mobilization, with a focus on expanding the tax base and reducing tax expenditures. Oil and gas-related revenues are likely to have a significant impact from 2023 onwards.
- **Current account deficit.** The baseline scenario assumes a current account deficit of 10.5 percent in 2020 and 11.3 percent in 2021, broadly in line with the 2nd PCI review. This reflects the protracted impact of the pandemic as well as the significant ongoing investments in the hydrocarbon sector. The current account deficit is projected to sharply decline in 2023 as oil and gas exports are expected to kick in.

9. The DSA assumes a financing mix consistent with a prudent borrowing strategy, emphasizing external financing on concessional terms in the near term but gradually increasing the share of domestic debt over time. Financing for the revised 2021 budget will be filled mostly with grants and concessional and semi-concessional loans (Text Figure 4), including from the World Bank, African Development Bank, and other bilateral partners. IMF financing under the SCF/SBA will reduce the need for Senegal to borrow on the regional market in 2021 and 2022. Beyond that, the authorities' medium-term debt strategy targets a gradual shift away from external financing denominated in U.S. dollars to reduce foreign exchange risk and a greater reliance on the regional CFAF market. Accordingly, the DSA assumes that domestic borrowing will increase over time and account for an average of about 35 percent of total financing over 2022–26, from about 18 percent currently. External commercial borrowing (Eurobonds) is assumed to gradually increase over the long term (with new issuances equivalent to the stock of maturing Eurobonds plus 20-30 percent) as Senegal shifts away from concessional external borrowing to access the market more regularly.

Text Table 3. Senegal: Evolution of Selected Macroeconomic Indicators

	2021	2022	2023	2024	Med. term[1]	Long term[2]
Real GDP growth						
Current DSA	3.7	5.5	10.8	6.1	6.3	5.2
Previous DSA[3]	5.2	6.0	11.9	8.7	6.2	5.1
Overall fiscal deficit (percent of GDP)[4]						
Current DSA	6.6	4.9	3.2	3.1	4.1	2.7
Previous DSA	5.8	5.2	4.0	3.5	4.8	2.7
Current account deficit (percent of GDP)						
Current DSA	11.3	10.5	5.5	3.2	6.9	3.2
Previous DSA	10.7	10.6	4.6	3.6	7.4	1.3
Exports of goods and services (percent of GDP)						
Current DSA	19.8	22.8	31.7	36.0	29.2	32.6
Previous DSA	22.3	24.1	32.7	34.1	28.1	32.2

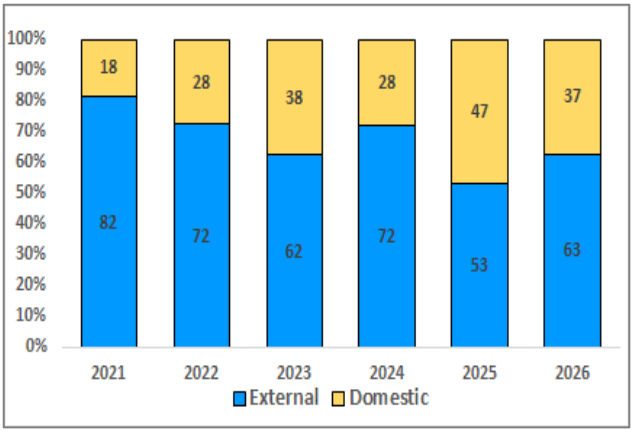
[1] Defined as the first 5 years of the projection period (2021-2025). For previous DSA 2020-2024.
[2] Defined as the last 16 years of the projection period (2026-2041). For previous DSA 2025-2040.
[3] 2nd PCI Review (Jan 2021)
[4] Overall fiscal deficit of General Government and Public Sector.

Source: IMF staff estimates and projections.

Text Figure 4. Senegal: 2021 Borrowing Plan and Financing Assumptions

PPG Debt	CFAF (billions)
Central Government	1415
External	
Concessional/Semi-concessional 1/	881
Commercial 2/	0
Domestic	334
Public/Parapublic enterprises (net)	250
Total - Public Sector	1665
Use of Financing	
Fiscal deficit (central government) 3/	796
Amortizations (central government)	619
Public/Parapublic enterprises (net)	250

1/ Debt with a positive grant element
2/ Debt without a positive grant element
3/ LFR budget deficit plus operational expenditures minus overfinancing from 2020



Source: Senegalese authorities, IMF staff estimates

10. The realism tools suggest that the proposed fiscal adjustment path is ambitious, but staff believe it is realistic under the circumstances. The assumed primary balance adjustment path of 4 percent of GDP over 2021-2023 is in the top quartile of the historical distribution for LICs (Figure 4). However, in the case of Senegal, a significant portion of the adjustment will come from the unwinding of temporary support measures introduced in response to the COVID-19 shock (estimated at 4.5 percent of GDP). Combined with the program's revenue-enhancing measures, staff believe that reaching the central government's 3 percent fiscal deficit target by 2023 is realistic, though not without risks. Separately, the projected economic growth rates in 2021 and 2022 are above the range of potential growth paths under various fiscal multipliers, but the COVID-19 pandemic and the related-recovery are not well-captured by the exercise. The DSA's realism tools do not signal any other signs of over-optimism in terms of large changes to investment or contributions to growth compared to previous DSAs (Figure 4).

COUNTRY CLASSIFICATION AND DETERMINATION OF STRESS TEST SCENARIOS

11. Senegal's debt carrying capacity is rated as strong. Based on data from the October 2020 World Economic Outlook database and the World Bank's 2019 Country Policy and Institutional Assessment (CPIA) score, Senegal's Composite Indicator (CI) is 3.19. This assessment affects the thresholds used to calculate the mechanical external debt risk ratings. Senega's CI score has decreased compared to the last vintage (3.30) due to lower Senegal and global growth and CPIA score, primarily reflecting the impact of the COVID-19 pandemic (Text Table 4).

Text Table 4. Senegal: Debt Carrying Capacity

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.753	1.45	45%
Real growth rate (in percent)	2.719	6.291	0.17	5%
Import coverage of reserves (in percent)	4.052	39.093	1.58	50%
Import coverage of reserves^2 (in percent)	-3.990	15.283	-0.61	-19%
Remittances (in percent)	2.022	10.261	0.21	6%
World economic growth (in percent)	13.520	2.928	0.40	12%
CI Score			3.19	100%
CI rating			Strong	

Source: IMF staff estimates

12. The standard stress tests have been applied, along with a market financing shock. The use of a tailored stress test for market financing reflects Senegal's outstanding Eurobonds. The test assumes a temporary increase in the cost of new commercial external borrowing (by 400 basis points), a nominal depreciation of the CFAF relative to the US dollar, and a shortening of maturities and grace periods. The results do not flag any areas of concern (Figure 5).

EXTERNAL DSA

13. External debt indicators mostly remain below their thresholds under the baseline scenario, with the exception of a temporary one-off breach. In terms of the DSA's solvency metrics, public external debt is projected to peak at 57 percent of GDP in 2021 before reversing as the recovery and fiscal consolidation take hold (Table 1). The present value of debt to exports breaches the risk threshold in the first year of the baseline before steadily declining, aided by the anticipated exports from the hydrocarbon sector in 2023 and beyond. As a short-lived one-year breach, this is ignored for the DSA risk ratings. In terms of the liquidity indicators, the ratio of external debt service to exports hit 24 percent in 2020 but is projected to gradually decline, averaging about 16 percent over the medium term. Debt service to revenues is also projected to hover around 15 percent over the same horizon.

14. Three of the four external debt indicators breach their threshold under the sensitivity analysis. The most extreme shock is a shock to exports in the case of the present value of debt to GDP, debt-to-exports and debt service-to-exports ratios. Other breaches occur for the same three risk indicators

under different scenarios as well (Table 3). The debt service-to-revenues indicators does not show any breaches of the thresholds after excluding the temporary one-off breach under the combination shock. Overall, these results point to potential vulnerabilities under adverse conditions, including a protracted or multi-speed global recovery that would hit Senegal's main export markets, or in case delays were to occur in the development of the hydrocarbon sector. Separately, the difference between the baseline and the historical scenario largely reflects the structural change implied by the anticipated oil and gas exports on the non-interest current account deficit (averaging 2.2 percent of GDP in the baseline against 6 percent in the historical scenario).

OVERALL RISK OF PUBLIC DEBT DISTRESS

15. The present value of public debt to GDP remains below its threshold under the baseline scenario. Total public debt is projected to peak at about 71 percent of GDP in 2021 before gradually declining (Table 2). The present value of debt to revenues is also projected to gradually decline over time. Debt service is projected to average about 24 percent of total revenues and grants over the next five years, which represents a significant portion of future fiscal revenues.

16. Stress tests indicate that Senegal is most vulnerable to a growth shock. Under such a shock (which simulates a growth rate of 2.4 percent in 2022–23), all three public debt indicators would be set on an explosive growth path. While this is an extreme shock (sweeping away part of the anticipated post-COVID recovery and the positive growth impact of oil and gas production), this underscores the importance of reforms to strengthen Senegal's medium-term growth potential combined with prudent fiscal deficits supported by enhanced domestic resource mobilization efforts.

RISK RATING AND VULNERABILITIES

17. Senegal remains at moderate risk of external debt distress, with limited capacity to absorb shocks in the near term (Figure 6). Heightened uncertainty over the global economic outlook suggests the need for a prudent approach that emphasizes macroeconomic stability and fiscal discipline, reforms to contain fiscal risks, and further efforts to strengthen debt management capacity.

18. Senegal's overall risk of debt distress also remains moderate. Still, the relatively high debt service levels throughout the forecast period reduce the space for public expenditures in support of a robust and inclusive economic recovery. The authorities should prioritize further efforts to mobilize additional revenues and seek out concessional borrowing where possible in the near term.

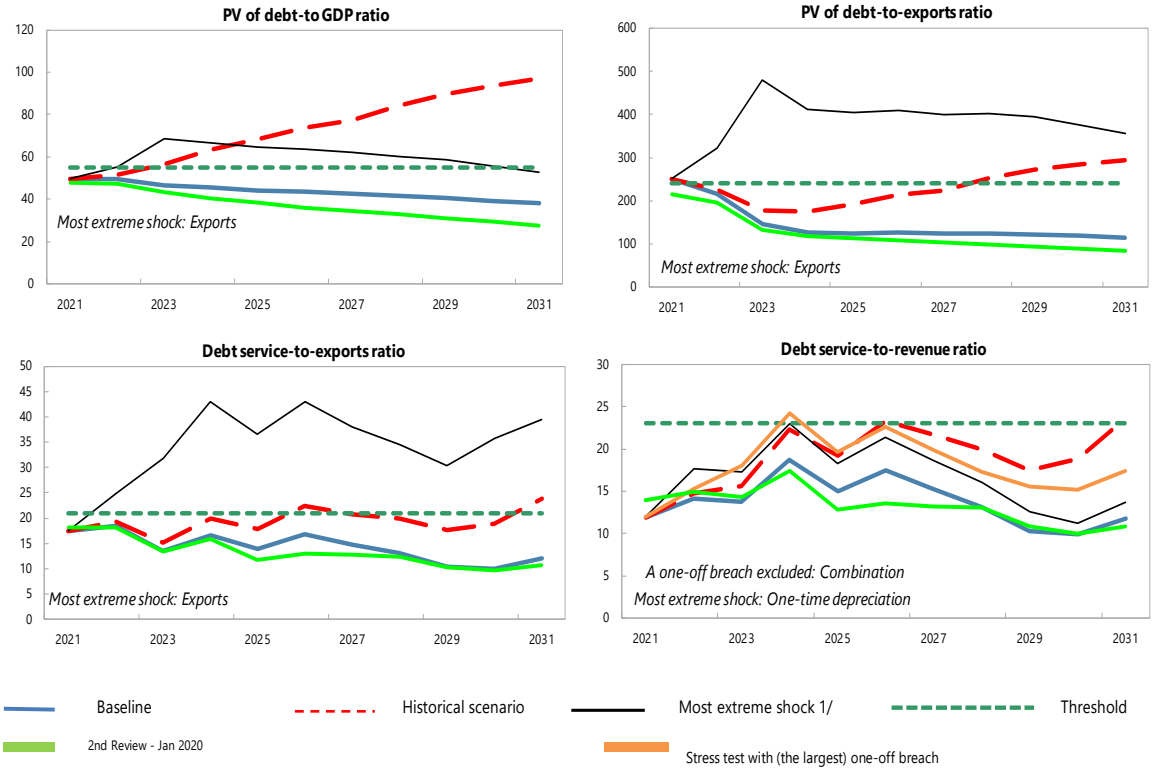
19. There are significant risks to the assessment. The outlook depends primarily on how quickly the COVID-19 pandemic can be contained in Senegal and its major trading partners. A rapid vaccination campaign would allow the government to ease restrictions on economic activity, supporting economic growth. By contrast, a slower recovery due a prolonged outbreak or delays in obtaining an adequate vaccine supply could further lower economic prospects. A quicker recovery in advanced economies could result in higher global interest rates, making market access more expensive for Senegal as it looks to rollover forthcoming Eurobond repayments. Likewise, higher oil prices resulting from a quicker global

recovery could result in higher budgetary costs due to domestic fuel subsidies. Over the medium term, sustained higher oil prices would increase the profitability of the new hydrocarbon projects, thereby strengthening the external and fiscal outlook. However, delays in hydrocarbon production would have an adverse impact on growth, revenues, and debt sustainability. Natural disasters, security concerns, and the potential for socio-political unrest are also risks.

AUTHORITIES' VIEWS

20. The authorities broadly share staff's assessment. They agree that risks to debt distress are moderate and that there is limited space to absorb shocks in the near term. Accordingly, the authorities are committed to contain risks to the debt path by adhering to the agreed debt ceiling, limiting non-concessional borrowing solely to projects with high prospective return, and implementing a medium-term debt strategy that seeks to mitigate exchange rate risks. This will be supported by efforts to strengthen the operational capacity of the CNDP and progressively improve their capacity to monitor debt-related developments in publicly and publicly owned enterprises.

Figure 1. Senegal: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2021–31



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	Yes	
Natural disaster	n.a.	n.a.
Commodity price	n.a.	n.a.
Market financing	No	No

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

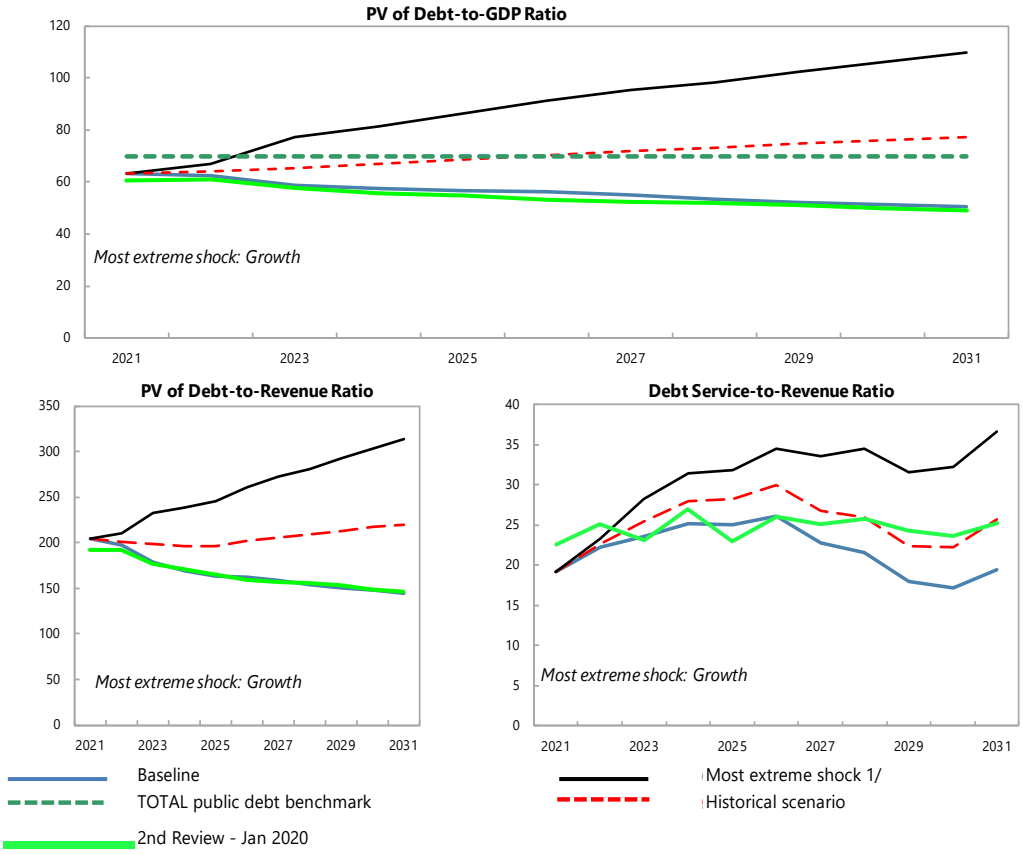
Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	3.8%	3.8%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	17	17
Avg. grace period	6	6

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Figure 2. Senegal: Indicators of Public Debt Under Alternative Scenarios, 2021–31



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	64%	64%
Domestic medium and long-term	33%	33%
Domestic short-term	4%	4%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	3.8%	3.8%
Avg. maturity (incl. grace period)	17	17
Avg. grace period	6	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	4.0%	4.0%
Avg. maturity (incl. grace period)	5	5
Avg. grace period	3	3
Domestic short-term debt		
Avg. real interest rate	3.1%	3.1%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

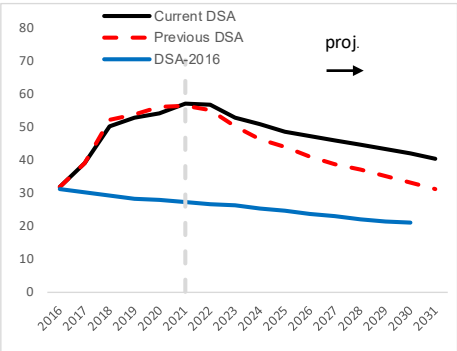
Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

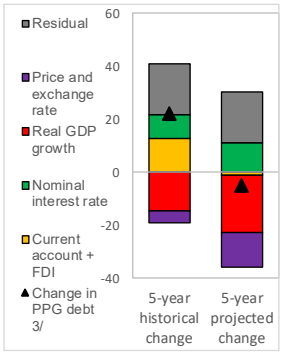
Figure 3. Senegal: Driver of Debt Dynamics—Baseline Scenario, 2016–31

External debt

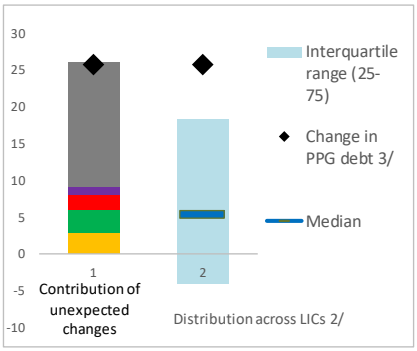
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

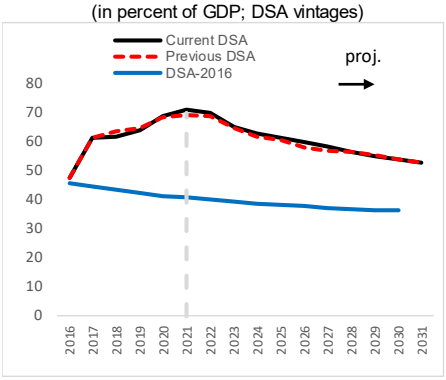


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

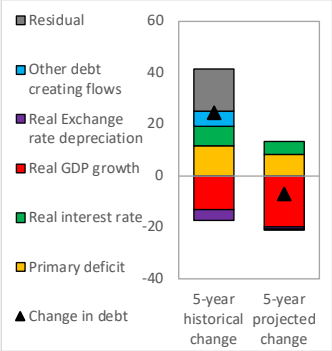


Public debt

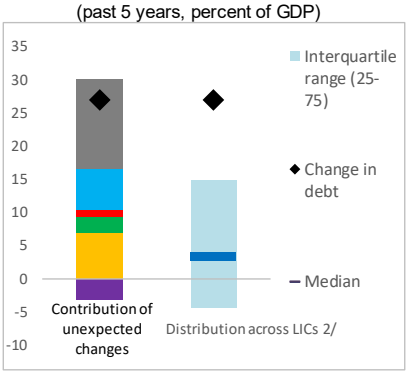
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



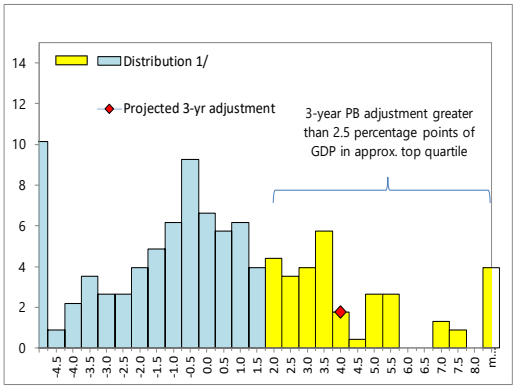
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

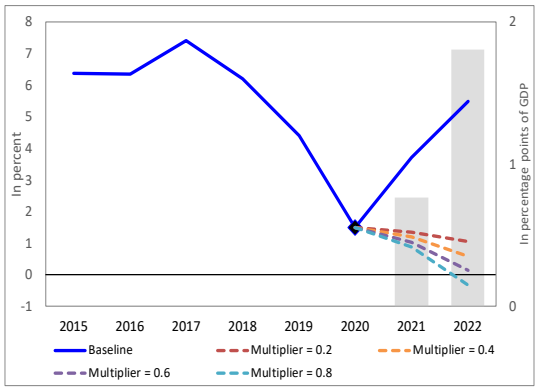
Figure 4. Senegal: Realism Tools, 2015–26

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



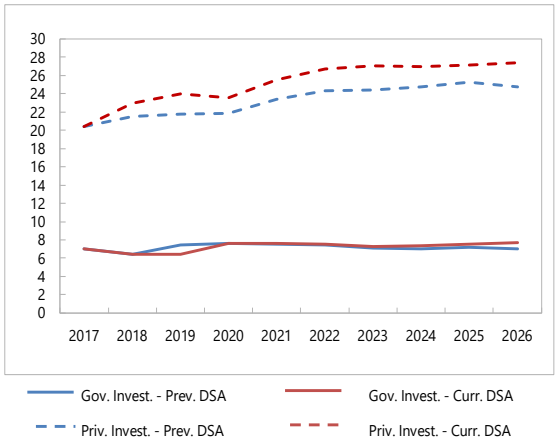
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)



Contribution to Real GDP growth
(percent, 5-year average)

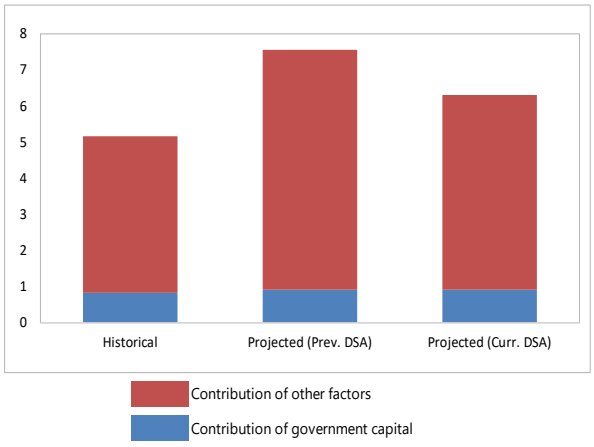
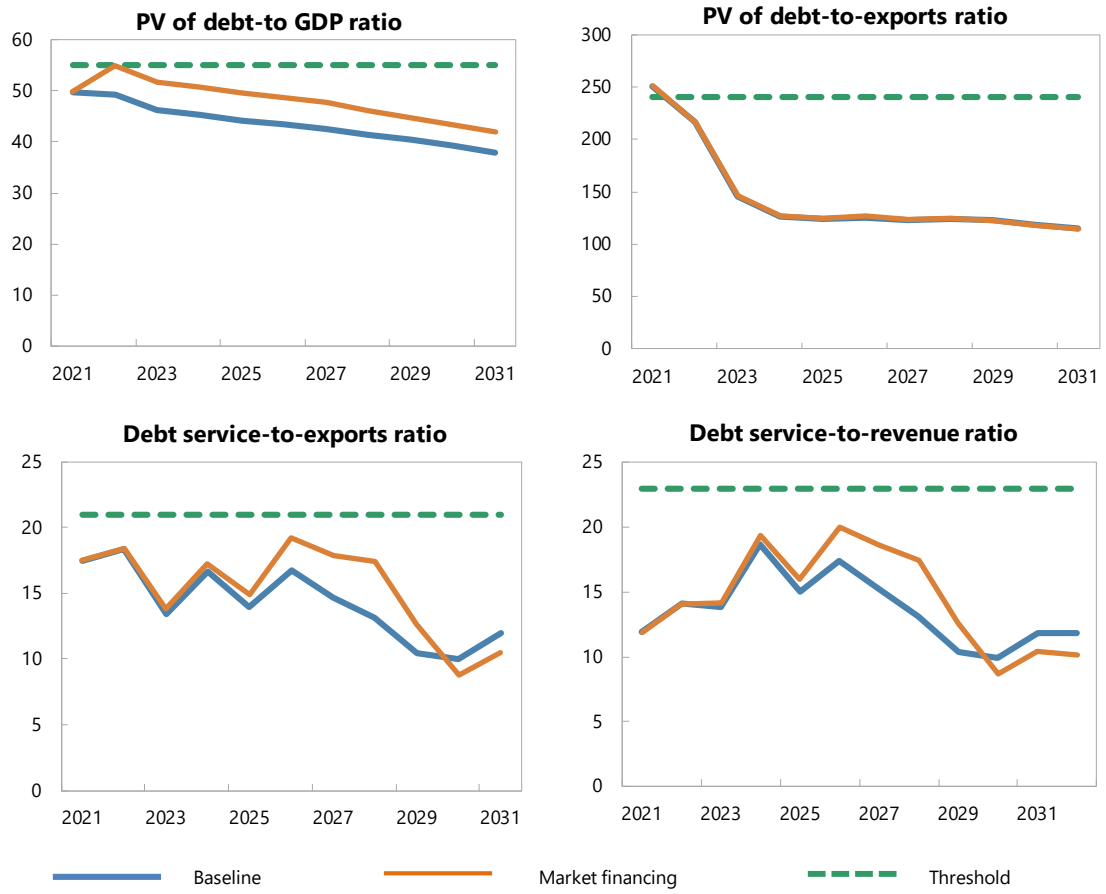


Figure 5. Senegal: Market-Financing Risk Indicators, 2021–31

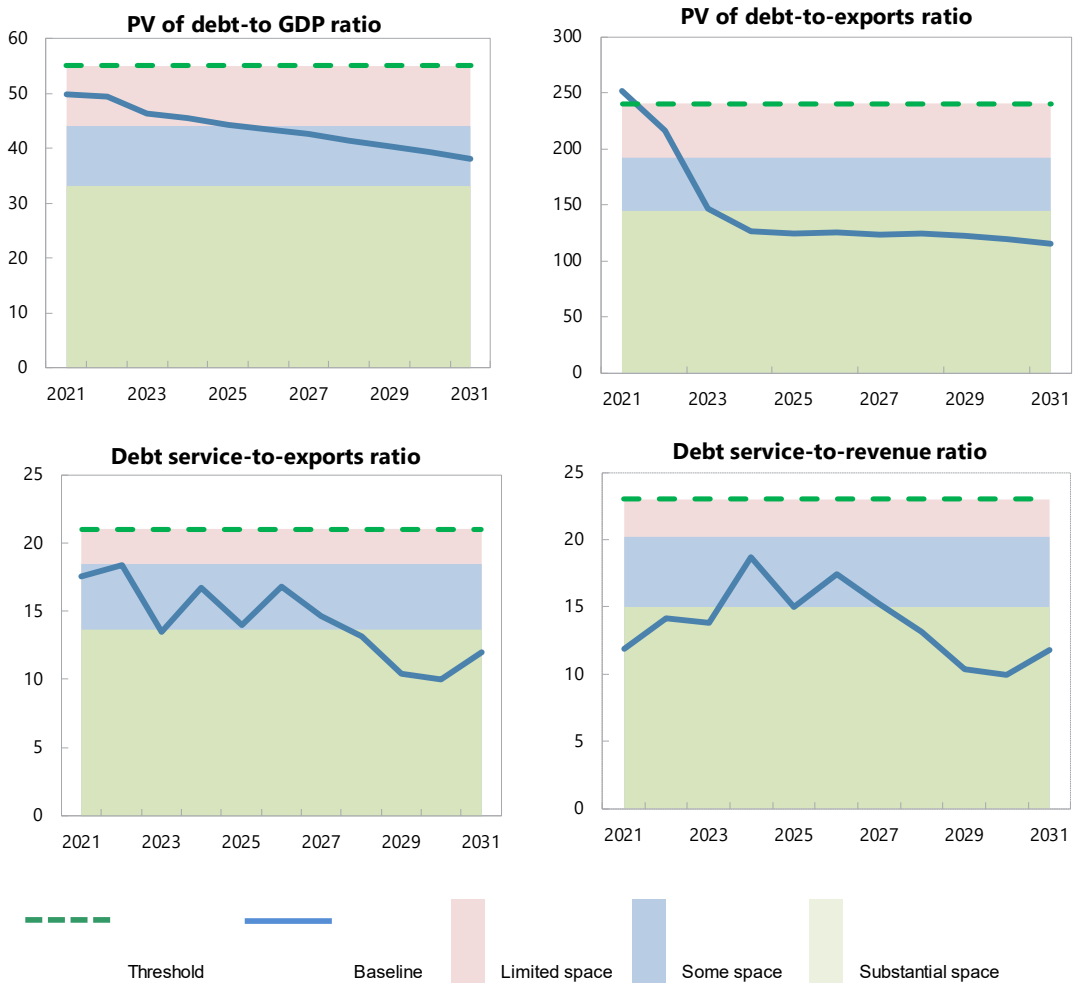
	GFN 1/	EMBI 2/
Benchmarks	14	570
Values	10	407
Breach of benchmark	No	No
Potential heightened liquidity needs	Low	

1/ Maximum gross financing needs (GFN) over 3-year baseline projection horizon.
2/ EMBI spreads correspond to the latest available data.



Sources: Country authorities; and staff estimates and projections.

Figure 6. Senegal: Qualification of the Moderate Risk Category, 2021–31[1]



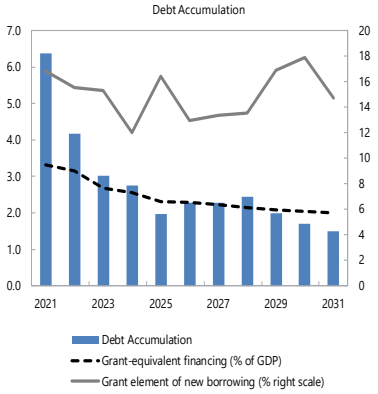
Sources: Country authorities; and staff estimates and projections.

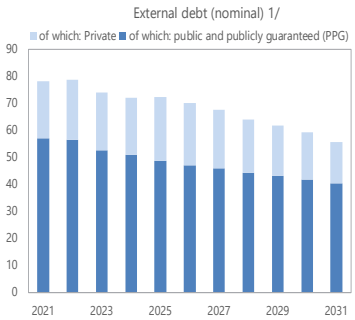
1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.

Table 1. Senegal: External Debt Sustainability Framework, Baseline Scenario, 2018–41

	Actual			Projections								Average 8/	
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
External debt (nominal) 1/	**70.0**	**73.9**	**75.4**	**78.0**	**78.8**	**74.1**	**72.1**	**72.2**	**70.1**	**55.8**	**24.2**	**59.5**	**68.5**
of which: public and publicly guaranteed (PPG)	**50.1**	**52.8**	**54.1**	**56.9**	**56.6**	**52.6**	**50.8**	**48.6**	**47.1**	**40.4**	**24.8**	**36.5**	**48.0**
Change in external debt	8.2	3.9	1.4	2.7	0.7	-4.7	-2.0	0.1	-2.2	-3.4	-2.8		
Identified net debt-creating flows	**-0.3**	**3.3**	**0.6**	**3.2**	**1.4**	**-6.6**	**-5.4**	**-4.3**	**-3.4**	**-2.0**	**-1.4**	**2.7**	**-2.6**
Non-interest current account deficit	**6.8**	**5.5**	**8.3**	**9.4**	**8.2**	**3.3**	**0.8**	**1.4**	**2.2**	**1.8**	**1.5**	**6.0**	**3.0**
Deficit in balance of goods and services	15.1	14.3	17.7	19.0	15.2	8.4	5.5	5.9	4.7	1.8	1.1	**14.8**	**6.1**
Exports	23.6	25.0	19.3	19.8	22.8	31.7	36.0	35.6	34.5	33.0	32.8		
Imports	38.8	39.3	37.0	38.8	38.0	40.1	41.5	41.6	39.2	34.7	33.8		
Net current transfers (negative = inflow)	-8.9	-9.0	-9.6	-9.6	-9.0	-8.7	-8.6	-8.5	-8.5	-8.1	-7.6	**-9.6**	**-8.5**
of which: official	-0.3	0.1	-1.3	-0.5	-0.5	-0.4	-0.3	-0.1	-0.1	-0.1	-0.3		
Other current account flows (negative = net inflow)	0.6	0.1	0.2	0.0	2.0	3.6	3.9	3.9	6.0	8.1	8.1	**0.8**	**5.4**
Net FDI (negative = inflow)	**-3.4**	**-4.3**	**-5.8**	**-5.6**	**-5.2**	**-4.5**	**-4.4**	**-4.5**	**-4.5**	**-2.8**	**-2.8**	**-2.6**	**-4.1**
Endogenous debt dynamics 2/	**-3.7**	**2.1**	**-1.9**	**-0.5**	**-1.7**	**-5.4**	**-1.8**	**-1.1**	**-1.2**	**-1.0**	**-0.1**		
Contribution from nominal interest rate	2.0	2.7	2.2	1.9	2.2	2.2	2.4	2.5	2.4	1.9	1.2		
Contribution from real GDP growth	-3.5	-3.1	-1.0	-2.5	-3.9	-7.5	-4.2	-3.6	-3.6	-2.9	-1.3		
Contribution from price and exchange rate changes	-2.2	2.5	-3.1	...	...	...	...	...	...	...	...		
Residual 3/	**8.6**	**0.6**	**0.9**	**-0.5**	**-0.7**	**1.9**	**3.5**	**4.4**	**1.2**	**-1.4**	**-1.4**	**0.7**	**0.8**
of which: exceptional financing	1.1	-1.1	1.3	-0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**49.8**	**49.7**	**49.4**	**46.4**	**45.4**	**44.1**	**43.4**	**38.0**	**23.3**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**257.7**	**251.2**	**216.6**	**146.0**	**126.1**	**123.8**	**125.7**	**115.1**	**71.0**		
PPG debt service-to-exports ratio	**15.9**	**13.3**	**24.2**	**17.5**	**18.4**	**13.4**	**16.7**	**14.0**	**16.8**	**11.9**	**8.6**		
PPG debt service-to-revenue ratio	**13.5**	**11.2**	**16.4**	**11.9**	**14.1**	**13.8**	**18.7**	**15.0**	**17.4**	**11.8**	**8.4**		
Gross external financing need (Billion of U.S. dollars)	1.8	1.4	2.0	2.2	2.6	1.4	1.3	1.2	2.0	2.4	2.6		
Key macroeconomic assumptions													
Real GDP growth (in percent)	6.2	4.4	1.5	3.7	5.5	10.8	6.1	5.4	5.3	5.3	5.0	**4.6**	**5.8**
GDP deflator in US dollar terms (change in percent)	3.7	-3.5	4.3	8.8	3.5	2.0	1.9	1.9	1.5	1.9	2.2	**0.0**	**2.7**
Effective interest rate (percent) 4/	3.6	3.9	3.2	2.9	3.1	3.1	3.5	3.7	3.6	3.5	4.7	**2.7**	**3.4**
Growth of exports of G&S (US dollar terms, in percent)	19.2	6.7	-18.3	15.7	25.7	57.4	22.7	6.3	3.6	7.2	9.7	**4.6**	**15.0**
Growth of imports of G&S (US dollar terms, in percent)	19.6	2.3	-0.5	18.5	6.9	19.2	11.9	7.5	0.9	9.8	9.4	**6.4**	**8.2**
Grant element of new public sector borrowing (in percent)	...	...	...	16.9	15.5	15.3	12.0	16.4	13.0	14.7	16.9	**...**	**15.0**
Government revenues (excluding grants, in percent of GDP)	27.7	29.6	28.6	29.1	29.7	30.9	32.1	33.2	33.2	33.3	33.5	**20.4**	**32.2**
Aid flows (in Billion of US dollars) 5/	1.0	1.5	1.7	0.7	0.7	0.7	0.7	0.7	0.8	1.0	1.7		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	3.3	3.1	2.7	2.6	2.3	2.3	2.0	1.6	**...**	**2.4**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	32.8	35.3	37.3	30.3	36.8	30.7	37.0	49.4	**...**	**35.2**
Nominal GDP (Billion of US dollars)	23	23	25	28	30	34	37	40	43	62	125		
Nominal dollar GDP growth	10.2	0.8	5.9	12.9	9.2	13.0	8.2	7.3	6.9	7.2	7.2	**4.5**	**8.7**
Memorandum items:													
PV of external debt 7/	...	...	71.1	70.8	71.6	67.8	66.7	67.7	66.3	53.3	22.7		
In percent of exports	...	...	368.1	357.8	314.1	213.7	185.3	190.1	192.1	161.6	69.3		
Total external debt service-to-exports ratio	19.1	18.7	29.8	21.3	23.6	16.8	20.1	17.3	20.2	14.9	10.4		
PV of PPG external debt (in Billion of US dollars)			12.3	13.9	15.0	15.9	16.9	17.6	18.5	23.4	29.0		
(PVt-PVt-1)/GDPt-1 (in percent)				6.4	4.2	3.0	2.7	2.0	2.3	1.5	0.0		
Non-interest current account deficit that stabilizes debt ratio	-1.4	1.5	6.8	6.7	7.5	8.0	2.8	1.3	4.4	5.2	4.3		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	Yes





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

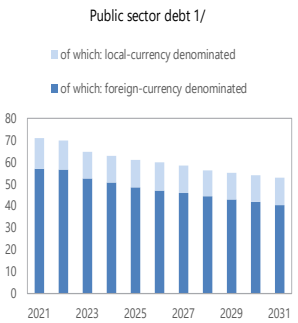
7/ Assumes that PV of private sector debt is equivalent to its face value.

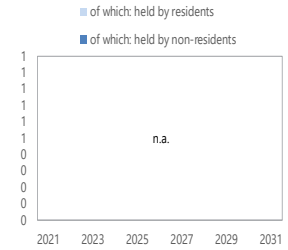
8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 2. Senegal: Public Sector Debt Sustainability Framework, Baseline Scenario, 2018–41

	Actual			Projections								Average 6/	
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
Public sector debt 1/	**61.5**	**63.8**	**68.7**	**70.9**	**69.9**	**64.9**	**62.8**	**61.1**	**59.9**	**52.8**	**44.6**	**49.4**	**60.6**
of which: external debt	**50.1**	**52.8**	**54.1**	**56.9**	**56.6**	**52.6**	**50.8**	**48.6**	**47.1**	**40.4**	**24.8**	**36.5**	**48.0**
Change in public sector debt	0.4	2.3	4.8	2.2	-1.0	-5.0	-2.0	-1.7	-1.2	-1.1	-0.6		
Identified debt-creating flows	**3.2**	**3.0**	**1.8**	**2.7**	**-0.6**	**-4.9**	**-2.0**	**-1.7**	**-1.3**	**-1.1**	**-0.6**	**2.6**	**-1.4**
Primary deficit	**1.8**	**2.4**	**4.8**	**4.0**	**2.2**	**0.8**	**0.7**	**0.4**	**0.5**	**0.5**	**0.5**	**2.6**	**1.0**
Revenue and grants	29.7	31.3	30.9	31.1	31.8	32.8	33.8	34.7	34.7	34.7	34.7	**22.5**	**33.9**
of which: grants	2.0	1.6	2.3	1.9	2.1	1.9	1.7	1.5	1.5	1.4	1.3		
Primary (noninterest) expenditure	31.5	33.7	35.7	35.1	34.0	33.6	34.5	35.2	35.2	35.2	35.2	**25.2**	**34.9**
Automatic debt dynamics	**0.8**	**-0.1**	**-4.3**	**-1.3**	**-2.9**	**-5.8**	**-2.7**	**-2.1**	**-1.8**	**-1.5**	**-1.1**		
Contribution from interest rate/growth differential	-0.8	-1.3	0.2	-1.3	-2.9	-5.8	-2.7	-2.1	-1.8	-1.5	-1.1		
of which: contribution from average real interest rate	2.8	1.3	1.2	1.1	0.8	1.1	1.0	1.1	1.3	1.2	1.0		
of which: contribution from real GDP growth	-3.6	-2.6	-0.9	-2.5	-3.7	-6.8	-3.8	-3.2	-3.1	-2.7	-2.1		
Contribution from real exchange rate depreciation	1.5	1.2	-4.6	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.6**	**0.7**	**1.3**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.7**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (below-the-line operations)	0.6	0.7	1.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**-2.8**	**-0.7**	**3.0**	**-0.5**	**-0.4**	**-0.1**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**1.4**	**-0.1**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**61.3**	**63.3**	**62.6**	**58.6**	**57.4**	**56.7**	**56.2**	**50.4**	**43.1**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**198.5**	**203.9**	**197.0**	**178.7**	**169.8**	**163.1**	**161.7**	**145.0**	**124.0**		
Debt service-to-revenue and grants ratio 3/	**31.4**	**20.9**	**23.5**	**19.1**	**22.2**	**23.6**	**25.2**	**25.0**	**26.0**	**19.4**	**19.5**		
Gross financing need 4/	11.7	9.7	13.4	10.0	9.3	8.5	9.2	9.1	9.6	7.2	7.3		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	6.2	4.4	1.5	3.7	5.5	10.8	6.1	5.4	5.3	5.3	5.0	**4.6**	**5.8**
Average nominal interest rate on external debt (in percent)	4.1	3.7	3.8	3.1	2.8	2.9	3.0	3.3	3.4	3.7	3.7	**2.9**	**3.3**
Average real interest rate on domestic debt (in percent)	4.6	3.7	4.0	3.9	3.6	4.0	3.9	3.8	4.1	4.0	3.8	**4.7**	**3.8**
Real exchange rate depreciation (in percent, + indicates depreciation)	4.0	2.4	-8.7	...	...	...	...	...	...	...	...	**1.1**	**...**
Inflation rate (GDP deflator, in percent)	-0.8	1.9	2.3	1.9	2.1	1.7	1.8	1.9	1.6	1.9	2.2	**1.3**	**1.9**
Growth of real primary spending (deflated by GDP deflator, in percent)	63.0	11.5	7.5	2.0	2.3	9.4	9.0	7.3	5.6	5.4	4.9	**11.5**	**5.7**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	1.4	0.1	0.0	1.8	3.3	5.9	2.7	2.1	1.8	1.5	1.1	**0.5**	**2.4**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**Yes**





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The entire public sector, including SOEs . Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 3. Senegal: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2021–31

(percent)

	Projections 1/										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of debt-to GDP ratio											
Baseline	50	49	46	45	44	43	43	41	40	39	38
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	50	52	**56**	**63**	**69**	**74**	**77**	**84**	**89**	**94**	**97**
A2. Alternative Scenario	48	47	43	40	39	36	34	33	31	29	28
B. Bound Tests											
B1. Real GDP growth	50	52	**55**	54	53	52	51	49	48	47	45
B2. Primary balance	50	50	50	49	47	47	46	45	44	43	41
B3. Exports	50	**55**	**68**	**67**	**65**	**64**	**62**	**60**	**59**	**56**	53
B4. Other flows 3/	50	53	54	52	51	50	49	48	46	44	43
B5. Depreciation	50	**62**	54	53	51	50	49	48	47	46	45
B6. Combination of B1-B5	50	**62**	**67**	**66**	**64**	**63**	**62**	**60**	**58**	**56**	53
C. Tailored Tests											
C1. Combined contingent liabilities	50	53	50	49	48	48	47	46	45	43	42
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	50	55	52	51	50	49	48	46	45	43	42
Threshold	55	55	55	55	55	55	55	55	55	55	55
PV of debt-to-exports ratio											
Baseline	**251**	217	146	126	124	126	123	124	122	119	115
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	**251**	227	178	176	192	214	223	**252**	**271**	**284**	**294**
A2. Alternative Scenario	**251**	215	150	129	126	129	123	124	119	111	103
B. Bound Tests											
B1. Real GDP growth	**251**	217	146	126	124	126	123	124	122	119	115
B2. Primary balance	**251**	220	157	135	133	135	133	135	133	129	125
B3. Exports	**251**	**322**	**479**	**412**	**405**	**410**	**400**	**403**	**395**	**375**	**356**
B4. Other flows 3/	**251**	234	169	145	143	145	142	143	140	135	129
B5. Depreciation	**251**	217	135	116	114	116	114	115	113	111	108
B6. Combination of B1-B5	**251**	**321**	175	**305**	**300**	**304**	**298**	**300**	**293**	**282**	**270**
C. Tailored Tests											
C1. Combined contingent liabilities	**251**	234	158	137	134	138	136	137	135	132	128
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	**251**	217	146	127	125	127	124	124	122	118	114
Threshold	240	240	240	240	240	240	240	240	240	240	240
Debt service-to-exports ratio											
Baseline	18	18	13	17	14	17	15	13	10	10	12
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	18	19	15	20	18	**22**	21	20	18	19	**24**
A2. Alternative Scenario	18	19	14	18	15	18	16	14	11	10	12
B. Bound Tests											
B1. Real GDP growth	18	18	13	17	14	17	15	13	10	10	12
B2. Primary balance	18	18	14	17	14	17	15	14	11	11	13
B3. Exports	18	**25**	**32**	**43**	**37**	**43**	**38**	**34**	**30**	**36**	**39**
B4. Other flows 3/	18	18	14	17	15	18	15	14	12	12	14
B5. Depreciation	18	18	13	16	14	16	14	13	10	9	11
B6. Combination of B1-B5	18	**23**	**29**	**36**	**30**	**36**	**32**	**29**	**26**	**26**	**29**
C. Tailored Tests											
C1. Combined contingent liabilities	18	18	14	17	14	17	15	14	11	11	12
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	18	18	14	17	15	19	18	17	13	9	11
Threshold	21	21	21	21	21	21	21	21	21	21	21
Debt service-to-revenue ratio											
Baseline	12	14	14	19	15	17	15	13	10	10	12
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	12	15	16	22	19	**23**	22	20	18	19	**24**
A2. Alternative Scenario	12	14	15	20	16	19	17	15	11	10	12
B. Bound Tests	12	14	15	20	16	19	17	15	11	10	12
B1. Real GDP growth	12	15	16	22	18	21	18	16	12	12	14
B2. Primary balance	12	14	14	19	16	18	16	14	11	11	13
B3. Exports	12	14	15	22	18	20	18	16	14	16	18
B4. Other flows 3/	12	14	14	20	16	18	16	14	12	12	14
B5. Depreciation	12	18	17	23	18	21	19	16	13	11	14
B6. Combination of B1-B5	12	15	18	**24**	20	23	20	17	16	15	17
C. Tailored Tests											
C1. Combined contingent liabilities	12	14	14	19	15	18	16	14	11	10	12
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	12	14	14	19	16	20	19	17	13	9	10
Threshold	23	23	23	23	23	23	23	23	23	23	23

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Senegal: Sensitivity Analysis for Key Indicators of Public Debt, 2021–31

(Percent of GDP, unless otherwise indicated)

	Projections 1/										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
	PV of Debt-to-GDP Ratio										
Baseline	63	63	59	57	57	56	55	53	52	51	50
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	63	64	65	67	69	**70**	**72**	**73**	**75**	**76**	**77**
A2. Alternative Scenario	60	61	58	56	55	53	52	52	51	50	49
B. Bound Tests											
B1. Real GDP growth	63	67	**77**	**81**	**86**	**91**	**95**	**98**	**102**	**106**	**110**
B2. Primary balance	63	64	64	62	61	61	60	58	56	55	54
B3. Exports	63	68	**79**	**77**	**76**	**75**	**74**	**71**	69	67	64
B4. Other flows 3/	63	67	66	64	63	63	62	60	58	57	55
B5. Depreciation	63	**75**	67	63	60	57	53	50	46	43	40
B6. Combination of B1-B5	63	62	65	65	66	67	67	67	67	67	68
C. Tailored Tests											
C1. Combined contingent liabilities	63	69	65	63	62	62	61	59	57	56	55
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	63	63	59	58	57	57	55	53	52	51	50
TOTAL public debt benchmark	70	70	70	70	70	70	70	70	70	70	70
	PV of Debt-to-Revenue Ratio										
Baseline	204	197	179	170	163	162	159	154	151	148	145
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	204	201	198	197	197	202	206	210	213	217	220
A2. Alternative Scenario	19	25	28	30	25	26	25	23	18	17	19
B. Bound Tests											
B1. Real GDP growth	204	210	233	238	246	261	273	281	292	303	314
B2. Primary balance	204	201	194	184	177	175	171	166	163	159	156
B3. Exports	204	213	242	229	219	217	212	205	200	192	185
B4. Other flows 3/	204	210	201	190	183	181	177	171	167	163	158
B5. Depreciation	204	236	206	187	173	164	155	143	134	126	117
B6. Combination of B1-B5	204	195	196	192	189	192	193	192	193	193	194
C. Tailored Tests											
C1. Combined contingent liabilities	204	218	197	187	180	178	174	169	165	162	159
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	204	197	179	170	164	163	160	154	150	147	144
	Debt Service-to-Revenue Ratio										
Baseline	19	22	24	25	25	26	23	21	18	17	19
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	19	23	25	28	28	30	27	26	22	22	26
A2. Alternative Scenario	19	25	28	30	25	26	25	23	18	17	19
B. Bound Tests											
B1. Real GDP growth	19	23	28	31	32	34	34	35	32	32	37
B2. Primary balance	19	22	24	26	26	27	25	23	19	19	21
B3. Exports	19	22	24	28	27	28	25	24	21	23	25
B4. Other flows 3/	19	22	24	26	26	27	23	22	19	19	21
B5. Depreciation	19	23	27	30	28	30	26	24	20	18	21
B6. Combination of B1-B5	19	22	26	28	28	29	26	26	23	22	24
C. Tailored Tests											
C1. Combined contingent liabilities	19	22	25	26	26	29	26	23	19	18	21
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	19	22	24	26	26	28	26	26	20	16	18

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the benchmark.

2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

3/ Includes official and private transfers and FDI.